KESTREL

The drone security analyst that never blinks

Test and Evaluation Report

What was measured, how, and where the numbers are unflattering

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8/8 scenarios · 100 tests · F_1 1.00

Every figure in this document is generated from a measured run. Re-running the evaluation suite regenerates the tables; no number is transcribed by hand.

FlytBase AI Engineer assignment

Contents

1 How to reproduce everything

```
uv run pytest -q                          # 90 tests, no network, no API key
uv run python scripts/run_evals.py        # scenarios, retrieval, chaos, leaderboard
uv run python scripts/bench_gate.py       # gate efficiency by context
uv run python scripts/probe_models.py     # endpoint reachability
uv run python report/generate_tables.py # regenerate every table in this document
```

Every table here is generated from the JSON those commands write. Nothing is transcribed by hand, so a stale claim becomes visible rather than persisting.

2 Unit and integration tests

Suite	Tests	What breaks without it
Site geometry	11	Zones silently overlap, mislabelling every detection inside them
Projection	11	A sign error puts detections in the wrong zone, with no symptom
Storage, ledger, memory	20	Tamper detection stops working; entity histories fabricate
Rules and triage	26	Loitering stops firing; the stray dog raises a breach
Ask KESTREL	22	An agent launches a drone unapproved, or cites a frame that does not exist
Total	**90**	

2.1 The two assertions that matter most

Everything else is ordinary hygiene. These two guard failures that would discredit the system, and both are the kind of property that silently stops being true.

2.1.1 No gated tool executes without approval

```
for tool in gated_tools:
    result = await registry.invoke(tool.name, {}, approved=False)
    assert result["requires_confirmation"] is True

# And the agent's own loop cannot even pass the flag:
assert "approved=True" not in inspect.getsource(AskKestrel.ask)
```

A prompt-based guard would pass a casual review while being one jailbreak from irrelevance. This is enforced by the code path.

2.1.2 Fabricated citations are caught

```
# The model is scripted to cite a frame that does not exist.
turn = await agent.ask("what happened?")
assert turn.verified is False
assert "frm_totally_made_up_00001" in turn.verification_note
```

The complement matters equally: an explanation or a refusal legitimately cites nothing and must *not* be penalised for it.

3 Scenario suite

Scenario suite

8/8 scenarios correct **1.00** precision **1.00** recall **1.00** F_1

True-negative scenarios (routine delivery, wildlife at the fence, shift change) are weighted equally in this suite. A security system that cries wolf is switched off, and then protects nothing.

Table 1: Scenario suite. Three of the eight expect silence and are weighted equally: a system that alerts on the delivery driver, the stray dog and the shift change is switched off within a week, and then protects nothing.

Scenario	Expected	Raised	Suppressed	Result
delivery-daytime	\{}textit{silence}	0	0	PASS
loiter-midnight	loitering	2	0	PASS
fence-breach-night	after-hours-vehicle, perimeter-breach, restricted-zone	5	0	PASS
repeat-visitor	after-hours-vehicle, baseline-anomaly	4	0	PASS
wildlife-false-positive	\{}textit{silence}	0	0	PASS
unattended-package	unattended-object	1	0	PASS
shift-change-busy	\{}textit{silence}	0	0	PASS
tailgating	after-hours-vehicle, tailgating	4	0	PASS

Three of eight expect silence, and are weighted equally

A stray dog at the fence at 03:00 has every surface signal of an intrusion — motion, night, high-priority zone, extended dwell. A system that alerts on it, on the delivery driver, and on the shift change is switched off within a week, and then protects nothing. The true-negative cases are the harder test.

Each scenario runs against an isolated database. Entity and baseline state leaking between them would let a later scenario inherit an earlier one's history, and the suite would pass for the wrong reason.

4 Gate efficiency

38.9% of frames never reached a model on real footage, **96.7%** on an idle patrol

Read these together, not separately

Real-footage figures are a floor, not a typical value: every available clip is a CV demo reel authored for continuous motion, which is close to worst-case for a gate. The idle context is a constructed static scene representing patrol idle time and is labelled as such.

Table 2: Gate efficiency by operational context. The gate is context-adaptive by design: it spends more where missing something is expensive.

Context	Analysed	Skipped	Skip rate
restricted zone, 02:00	40	20	33.3%
substation, 02:00	40	20	33.3%
yard, 14:00	33	27	45.0%
staff parking, 14:00	33	27	45.0%
parking, quiet clip, 14:00	42	18	30.0%
yard, quiet clip, 11:00	32	28	46.7%
synthetic-idle (constructed)	2	58	96.7%

A prediction that did not survive contact with data

The design plan asserted "∼94% of frames never reach a VLM". Measured on real footage it is 38.9%. The number is reported rather than retuned until it flattered.
Both figures are quoted together because either alone misleads. The lower rate in the restricted zone at 02:00 is *correct behaviour*, not a regression: the gate deliberately raises sensitivity in high-priority zones, at night, and outside a zone's declared hours. It spends more where missing something is expensive.

5 Model reachability

Table 3: Every vision model in the catalogue, probed directly. Signals counts how many of three salient facts (colour, vehicle, person) the model reported for the probe image.

Model	Best latency	Signals	Status
`nvidia/llama-3.1-nemotron-nano-vl-8b-v1`	0.86 s	2/3	PASS
`meta/llama-3.2-11b-vision-instruct`	1.28 s	3/3	PASS
`nvidia/nemotron-nano-12b-v2-vl`	29.21 s	3/3	PASS
`meta/llama-3.2-90b-vision-instruct`	57.60 s	3/3	PASS
`nvidia/vila`	—	—	*404*
`microsoft/phi-3-vision-128k-instruct`	—	—	*404*
`microsoft/kosmos-2`	—	—	*404*
`nvidia/neva-22b`	—	—	*404*
`adept/fuyu-8b`	—	—	*404*

Catalogued is not reachable

4 / 9 catalogued vision models were actually callable. The rest return `Function not found for account`. A model appearing in a provider's catalogue is not evidence that you can call it, and probing first changed four of five original model choices.

6 Retrieval

Table 4: Hybrid retrieval over one ingested session. Indicative rather than a benchmark — four queries, one session.

Query	Plan	Relevant	P@k	Latency
show me all person events	structured	10/10	1.00	2120 ms
anything at the substation	structured	4/4	1.00	2021 ms
someone in a high-visibility vest	visual	9/10	0.90	2118 ms
activity in the restricted core	structured	4/4	1.00	1882 ms

0.97 mean precision@k.

Indicative rather than a benchmark — four queries, one ingested session. It exists to show the fusion works and the generated query plan is sane, and it is described that way in its own output rather than presented as a headline result.

7 Resilience

Table 5: Fault injection. Two cases expect *failure*: a cassette miss must raise rather than silently reaching the network, and a camera ray above the horizon must return no position rather than a plausible wrong one.

Injected fault	Required behaviour	Result
no API key	falls back to replay	PASS
cassette miss in replay	raises rather than silently calling out	PASS
corrupt / noise frame	gate still returns a verdict	PASS
VLM unavailable	degrades to a low-confidence graph	PASS
telemetry dropout	gate handles a null telemetry	PASS
ray above horizon	returns no position rather than a wrong one	PASS

6/6 faults degraded gracefully.

Two cases expect *failure*, deliberately:

- **Cassette miss** must raise rather than silently reaching the network. A quiet fallthrough would make "runs offline" untrue in exactly the situation where it matters.
- **Ray above horizon** must return no position at all. A plausible fabricated coordinate would send a responder somewhere real and wrong.

8 What is not tested

Stated plainly

A testing document that only lists successes is not useful.

- **Caption quality is not scored against ground truth.** There is no human-labelled caption set, so "the model describes frames well" rests on a three-signal probe and on inspection, not a metric.
- **Re-identification accuracy has no labelled set.** The tests assert the matching *logic* — same track merges, dissimilar appearance does not, people never merge with vehicles — not precision on a re-ID benchmark.
- **Projection is verified against analytic geometry, not survey data.** The mathematics is right; whether it matches a real site depends on calibration that does not exist here.
- **No load testing at fleet scale.** Throughput was not measured beyond a single session.
- **The frontend has smoke coverage only** — it builds and typechecks; there are no component tests.

9 Bugs found by tests, not by reading

Every one of these was invisible on inspection. Listing them is the most honest evidence that the suite is doing work.

1. **Gate reported 0% efficiency.** The heartbeat was time-only, and the demo compresses the site clock, so it fired every frame — the exact inverse of the architecture's central claim.
2. **`"cat"` is inside `"location"`.** Counterfactual keywords matched as substrings, so every alert titled "…at a sensitive location" was suppressed as wildlife.
3. **Scaling a confidence cannot express "this premise is false".** The stray-dog alert scaled to 0.2975 against a 0.25 threshold and still fired. Counterfactuals now split into contradicting and mitigating.
4. **Zone oscillation erased dwell.** Detections flicker between nested zones, resetting the clock. Loitering became undetectable in the highest-priority zone on the site. Symptom: zero alerts, no error.
5. **Label specificity.** The vision model says "sedan" where the rule says "car", and "vehicle" where it says "truck". Rules silently never matched.
6. **A zone with no declared hours was treated as always open**, disabling the after-hours rule at precisely the fence line and restricted core it exists to protect.